Institutional profiles under a common prudential framework

Evidence from Brazilian credit cooperatives and banks

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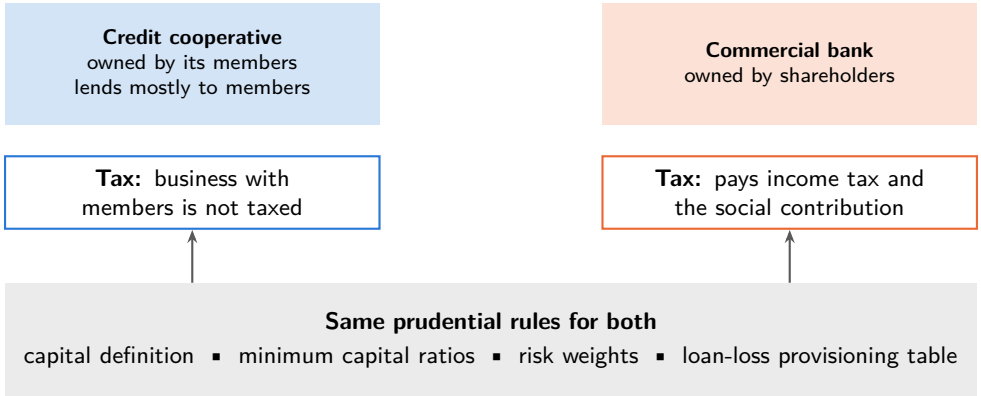
Organization of Brazilian Cooperatives (OCB System)

ICA CCR Europe Research Conference, Valencia, 4 November 2026

SistemaOCB

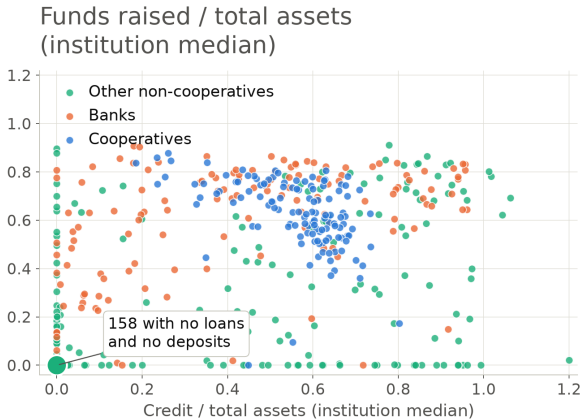
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Prudential framework



The common prudential framework is the selection criterion for the comparison.

Sample and comparison group



130 credit cooperatives

144 banks

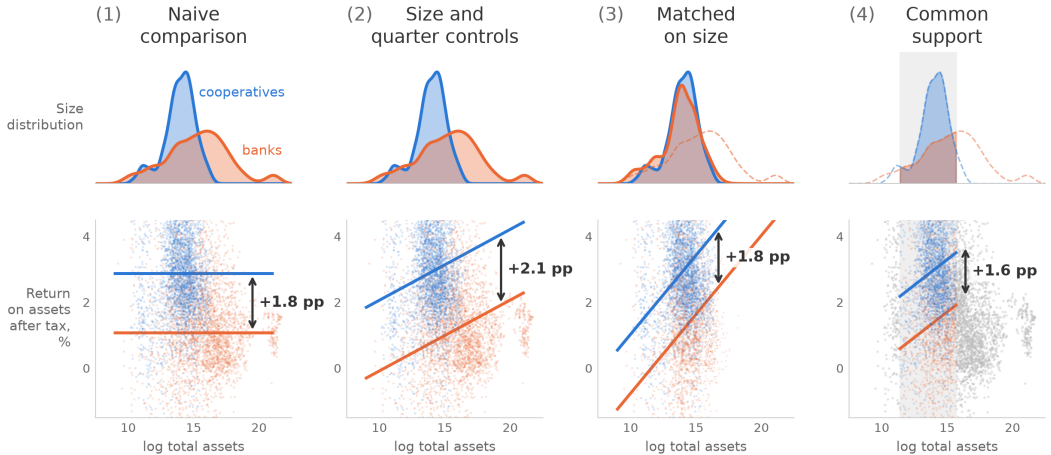
339 other non-coops

Every institution under the full Basel methodology, 2017 to 2024, from IF.data, the Central Bank of Brazil's public repository of quarterly filings. Cooperatives and banks: 7,234 institution-quarters. Central cooperatives are excluded.

Why compare to banks

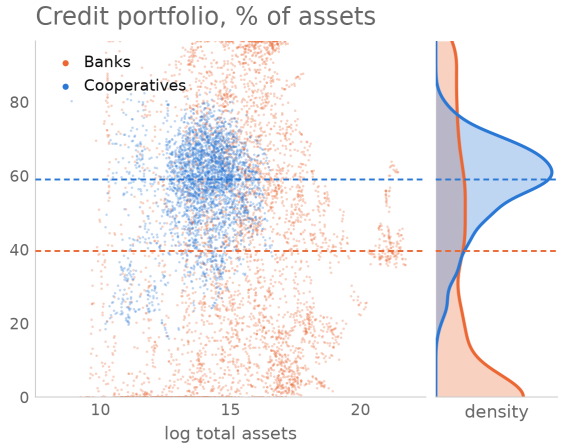
158 of the other non-cooperatives hold no loans and raise no funds; the other 181 are other lenders and non-bank institutions. Against them, the credit and funding differentials are mechanical. The broad group returns in Result 3.

Four specifications



Scatter: institution-quarters. Lines: the fitted specification.
The arrow is the cooperative differential in the after-tax return; $p < 0.001$.

Cooperatives lend more of their assets than banks of the same size



Institution-quarters; dashed lines: medians; outliers not shown.

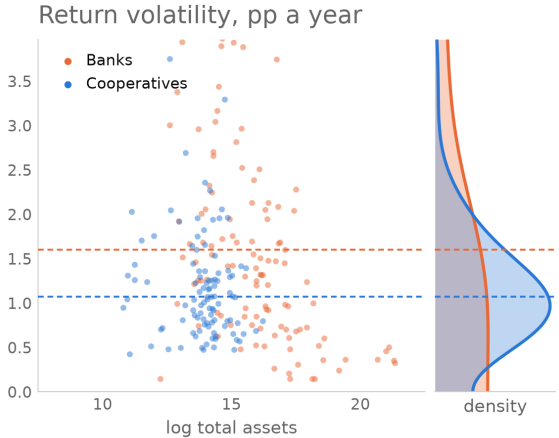
Specification	Differential	95% interval
(1) Naive comparison	+16.5 pp	11.2 to 21.9
(2) Size and quarter controls	+18.8 pp	12.9 to 24.8
(3) Matched on size	+17.0 pp	9.4 to 24.6
(4) Common support	+17.8 pp	10.7 to 24.9

Cooperatives hold 17 to 19 points more of their assets in loans than banks of the same size. Credit share rises with size and banks are larger, so the raw comparison slightly understates the gap.

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Cooperative returns are steadier



Institutions with at least eight quarters; dashed lines: medians; outliers not shown.

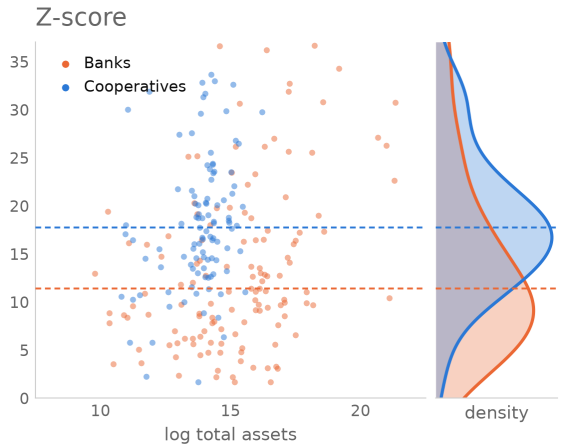
Specification	Differential	95% interval
(1) Naive comparison	−1.1 pp	−1.5 to −0.7
(2) Size controls	−1.9 pp	−2.3 to −1.5
(3) Matched on size	−1.8 pp	−2.2 to −1.3
(4) Common support	−1.7 pp	−2.2 to −1.2

Cooperative returns vary less over time: 1.1 points a year less in the raw comparison and 1.7 to 1.9 points less at equal size. Volatility falls with size and banks are larger, so the raw comparison understates the gap.

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Cooperatives sit further from insolvency



Institutions with at least eight quarters; dashed lines: medians; outliers not shown.

Specification	Differential	95% interval
(1) Naive comparison	+4.2	2.0 to 6.4
(2) Size controls	+6.3	4.2 to 8.4
(3) Matched on size	+6.1	3.6 to 8.7
(4) Common support	+6.2	3.4 to 9.0

$$z = \frac{\overline{\mathrm{ROA}} + E/A}{\sigma_{\mathrm{ROA}}}$$

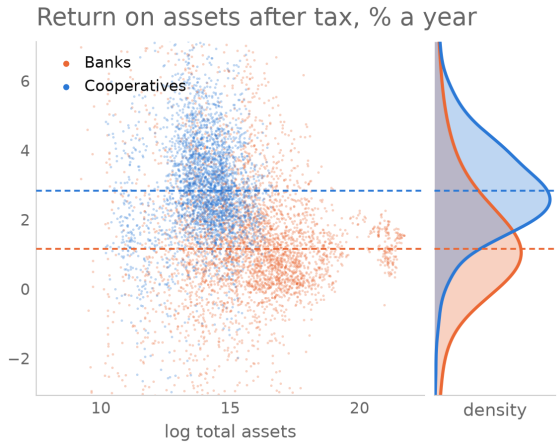
Mean return on assets plus equity over assets, over the standard deviation of the return, per institution: the number of standard deviations of return that would exhaust equity (Hesse and Čihák 2007).

Cooperatives sit further from insolvency than banks in every specification. Their equity ratio is lower at equal size, so the higher z-score comes mainly from steadier returns.

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Cooperatives earn more after tax

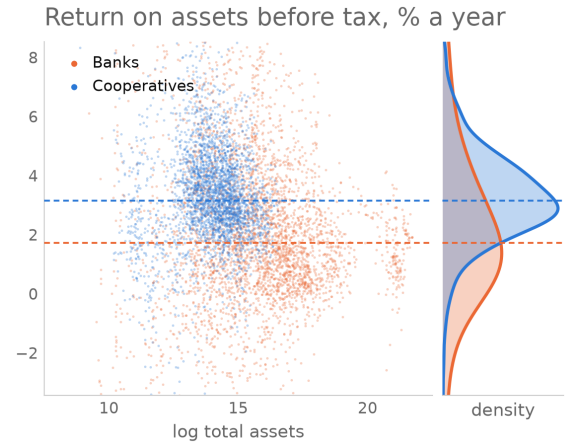


Institution-quarters; dashed lines: medians; outliers not shown.

Specification	Differential	95% interval
(1) Naive comparison	+1.8 pp	1.3 to 2.3
(2) Size and quarter controls	+2.1 pp	1.4 to 2.9
(3) Matched on size	+1.8 pp	1.2 to 2.4
(4) Common support	+1.6 pp	0.9 to 2.3

At equal size, cooperatives keep 1.6 to 2.1 points of assets more a year after tax than banks. The gap is about the same in all four specifications. The next slide shows the same comparison before tax.

Before tax, the average gap is smaller and less robust

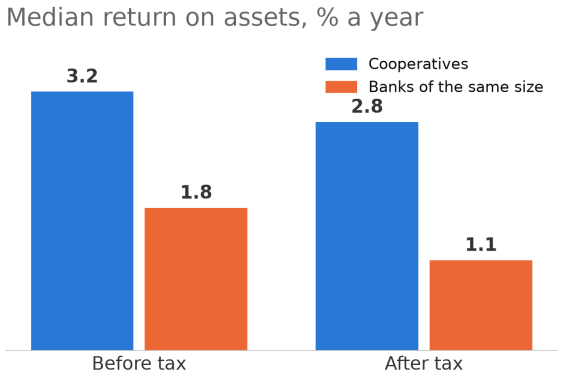


Institution-quarters; dashed lines: medians; outliers not shown.

Specification	Differential	95% interval
(1) Naive comparison	+1.2 pp	0.5 to 1.8
(2) Size and quarter controls	+1.3 pp	0.4 to 2.3
(3) Matched on size	+0.9 pp	0.0 to 1.8
(4) Common support	+0.6 pp	−0.4 to 1.6

Before tax, the gap is smaller and significant only in (1) and (2).

On the taxation of cooperatives



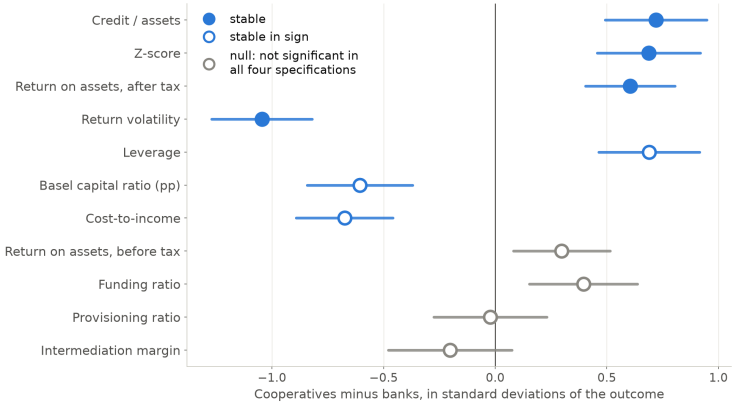
Medians with size and quarter controls (2); the tables on earlier slides show averages. After tax: net of tax and profit sharing.

For the typical cooperative, most of the after-tax gap is already there before tax. Three things set its surplus apart from a bank's profit:

- A credit cooperative lends to and takes deposits from its own owners. The surplus belongs to them, so the law does not treat it as profit earned from others. Operations with non-members are taxed.
- It is partly what members paid above cost. A larger surplus can mean higher prices to members, returned to them later.
- What is retained goes largely to mandatory reserves that cannot be distributed. It is the main way a cooperative builds capital.

Laws 5.764/1971 and 10.865/2004. Median effective tax rate in profitable quarters: banks 35%, cooperatives 1%.

Result 1. Eleven outcomes at comparable size



Cooperatives minus banks of comparable size, specification (2), in standard deviations of each outcome; 95 percent intervals.

We call a result stable when it holds in all four specifications and is not driven by extreme institutions.

Stable

Cooperatives lend more, earn more after tax and have steadier returns, sitting further from insolvency.

Stable in sign

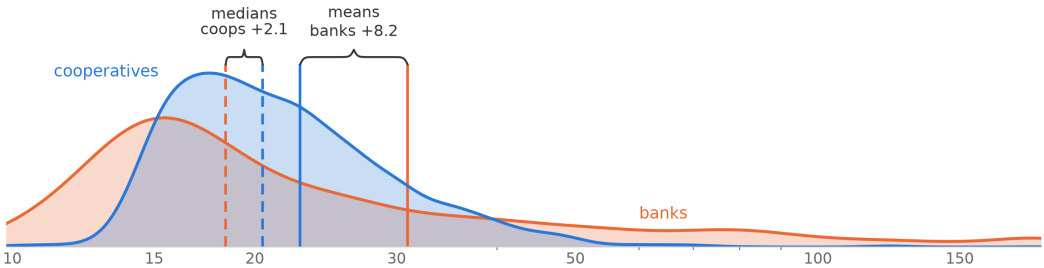
Capital, leverage and cost-to-income differ in the same direction everywhere, but a tail of banks drives the size of the gap.

Null

Before-tax return, funding, provisioning and margin.

Result 2. Capital: the average overstates the typical gap

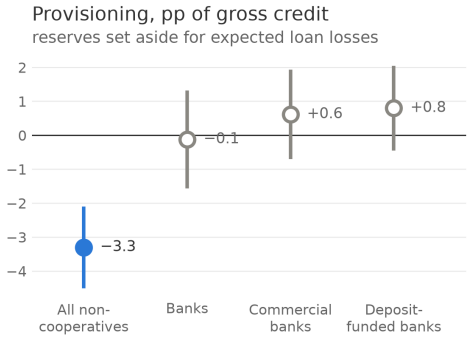
Basel capital ratio, % of risk-weighted assets (log scale)

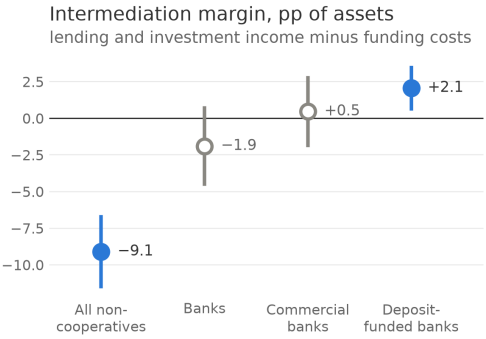


Institution-quarters, 2017 to 2024.

A tail of banks that are small or barely lend, with ratios above 50 percent, pulls the bank mean up. Considering typical institutions, cooperatives hold about as much capital as banks.

Result 3. Provisioning and margin depend on the comparison group

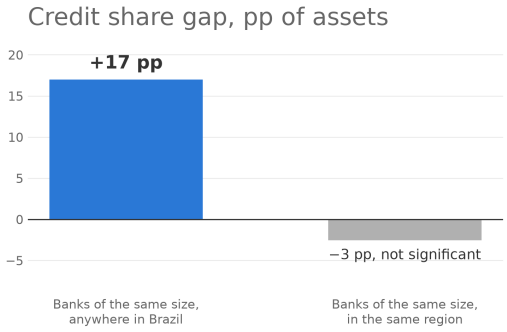




Cooperatives minus each comparison group, size and quarter controls (2), with 95 percent intervals. Filled: $p < 0.05$.

Against all non-cooperatives, cooperatives provision less and earn a thinner margin. That group includes non-bank lenders, such as consumer-finance companies, with high rates and heavy provisioning. Against any definition of a bank, neither holds.

An open question: region and the lending differential



Nationally

Cooperatives hold about 17 points more of their assets in loans than banks of the same size.

Within the same region

The gap disappears. Cooperatives are concentrated in the South, banks in the Southeast, and only about eighteen banks serve as controls.

Two readings

Either cooperatives lend more because of where they locate, with region as a mediator, or region explains the gap as a confounder. The data do not distinguish the two.

Specification: matched on size (3).

Brazil matches the literature except on profitability

	Theory expects	Brazil, this paper	Cross-country evidence
Capital	Less, in a narrower band	Less on average, narrower band; the typical cooperative holds about as much	Lower capitalisation: agrees
Return volatility	Lower	Lower	Lower: agrees
Profitability	No sign committed	Higher for the typical cooperative, before and after tax	Lower: differs
Provisioning, margin	No difference	None against banks	–

The difference on profitability is narrower than it looks: part of the after-tax gap is the tax exemption, the average before-tax gap is not established, and the sample holds the largest cooperatives.

Theory: Cook (1995), Fonteyne (2007). Cross-country: Hesse and Čihák (2007); Germany: Beck et al. (2009).

Summary

1. Profile

At the same size, cooperatives lend more, run leaner, earn more and have steadier returns. For the typical cooperative, most of the after-tax profit gap is already there before tax.

2. Capital

The typical cooperative holds about as much capital as the typical bank. The average gap comes from a bank tail that cooperatives do not have.

3. Comparison group

Lower provisioning and thinner margins appear only against a group that includes non-bank lenders. Against banks, no difference.

For regulators

A benchmark set on the average institution describes a lending cooperative poorly. The full distribution is the better guide for calibration. The results do not by themselves support lighter rules.

Limits

The estimates are conditional correlations at equal size. The sample is the largest tenth of Brazilian cooperatives. Code and data pipeline are public.

For discussion: how other frameworks treat cooperatives

Separate regime

Credit unions in the US, the UK and Ireland; savings cooperatives in Mexico. Own capital rules, often with lending limits. US credit unions: 7% net worth, no federal income tax.

Same rules, adapted capital

EU: member shares count as core capital, and a cooperative network can be supervised as one unit. Similar instruments for UK building societies and Australian mutual banks.

Same rules, tiers by size

Brazil, Indian urban cooperative banks, small banks in the EU and the UK. ECB: differences in treatment follow risk and complexity, whatever the legal form.

Tax varies most: cooperatives are exempt in Brazil and the US; Canada (2013) and India (2006) ended their tax breaks. EU Court of Justice: such exemptions are not necessarily State aid.

Open points for discussion

- The EU and the UK are preparing simpler rules for small banks, defined by size. In Brazil, cooperatives still differ from banks of the same size, so size alone may not separate the two.
- Regulators can recognise how cooperatives are built, with member shares as capital and mutual support within networks, without lowering capital ratios.
- Whether a tax exemption for cooperatives is a subsidy or reflects that their surplus belongs to the members.

Sources: NCUA; PRA; Central Bank of Ireland; CNBV; EU CRR Art. 10, 29, 113(7); ECB (2025); European Commission (2026); APRA; RBI; Basel III (2011); CJEU Paint Graphos (2011).